Conditional Wasserstein Autoencoders and Deep Semantic Representations for Time Series Modeling.

International Conference on Statistics and Data Science, Institute of Mathematical Statistics.

Seville, Spain, 16 dec. 2025.

Théo Guigue,
Françoise Seyte,
Roman Mestre
University of Montpellier
Montpellier Research in Economics

Summary

Context and Motivation

Central Question:

- How does the communication of European Central Bank, specifically through Forward Guidance (FG), impact systemic financial stress?

Goal of this Study:

- Build a continuous FG indicator based on the semantic content of ECB speeches.
- Compare different NLP models (ModernBERT, Mamba, etc.) to extract policy stance.
- Assess whether FG helps explain variations in systemic stress using conditional WAE-LSTM models.

MONTPELLIER RECHERCHE ECONOMIE

Data

Dataset	Period	Frequency	Description
ECB Text Corpus	Jan 2008 – Dec 2024	Monthly	Press releases, speeches and minutes scraped from ECB website; preprocessed into token sequences.
CISS Index	Jan 2008 – Dec 2024	Monthly	Composite Indicator of Systemic Stress (CISS) from Brunnermeier et al., capturing financial market stress.

Table: Summary of data sources used in the study

Encoders for Forward Guidance: Attention vs State-Space Dynamics

Goal: Encode long-horizon monetary policy texts while preserving slow-moving semantic signals.

Transformer-based encoder, ModernBERT (Warner et al., 2024):

$$\text{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \quad Q, K, V \in \mathbb{R}^{L\times d}$$

$$\mathcal{C}_{\text{time}} = \mathcal{O}(L^2), \quad \mathcal{C}_{\text{memory}} = \mathcal{O}(L^2)$$

ModernBERT improves scalability via:

RoPE, GeGLU, FlashAttention, local/global attention blocks

State-Space-based encoder, Mamba (Gu and Dao, 2024), Mamba-2 (Dao and Gu, 2024), Llamba (Wang et al., 2025): Continuous-time latent dynamics:

$$\frac{d\mathbf{y}(t)}{dt} = \mathbf{A}(x_t)\mathbf{y}(t) + \mathbf{B}(x_t)x_t$$

Discretized selective SSM:

$$\mathbf{y}_t = \exp(-\Delta t\, \mathbf{D}_t)\mathbf{y}_{t-1} + \mathbf{B}(x_t)x_t \quad \text{where} \quad \mathbf{A}_t = \mathrm{diag}(\alpha_t) = \exp(-\Delta t\, \mathbf{D}_t)$$

$$\mathbf{h}_t = \mathbf{W}_{\text{out}}\mathbf{y}_t, \quad \mathcal{C}_{\text{time}} = \mathcal{O}(L)$$

Key Insight:

Attention $\Rightarrow$ token-to-token alignment vs Mamba $\Rightarrow$ persistent semantic state

Semantic Axis Construction and Forward Guidance Scoring

Sentence embedding. Given a sentence $x = (t_1, \dots, t_L)$, the encoder produces token embeddings

$$H = [h_1, \dots, h_L]^\top \in \mathbb{R}^{L \times d}, \qquad h_i \in \mathbb{R}^d.$$

Sentence-level representation (mean pooling): $\mathcal{E}_x = \frac{1}{L} \sum_{i=1}^{L} h_i \in \mathbb{R}^d$ £

Semantic Forward Guidance axis. Using anchor sentences (expansive vs. restrictive generated by Gemini, Mistral 7B and Llama) we build a semantic embedding axis for discrimination:

$$\mathcal{V} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{E}_{x_i^+} - \frac{1}{M} \sum_{j=1}^{M} \mathcal{E}_{x_j^-}, \qquad \hat{\mathcal{V}} = \frac{\mathcal{V}}{\|\mathcal{V}\|}.$$

Projection and scoring. Each sentence embedding is projected into the forward guidance semantic axis using cosine similarity:

$$s_x = \cos(\mathcal{E}_x, \hat{\mathcal{V}}) = \frac{\mathcal{E}_x^\top \hat{\mathcal{V}}}{\|\mathcal{E}_x\|}.$$

Anchored normalization (score $\in [-1, 1]$):

$$z_x = \frac{s_x - \mu_{\text{anchor}}}{\sigma_{\text{anchor}}}, \quad (\mu_{\text{anchor}}, \sigma_{\text{anchor}}) = \mathrm{mean}, \mathrm{std}(\{s_{x_i^+}, s_{x_j^-}\}).$$

Encoding and latent space

We have multivariate time series $\{x_t \in \mathbb{R}^d\}_{t=1}^T$ and, optionally, an exogenous variable c_t. We then construct sliding windows of length L :
$X_i = \left[x_i,\ x_{i+1},\ \ldots,\ x_{i+L-1}\right] \in \mathbb{R}^{L\times d}, \quad C_i = c_{i+L-1} \in \mathbb{R}$, for $i = 1, \ldots, N$ with $N = T - L + 1$.
Each window X_i is encoded into a hidden representation by an LSTM encoder:

$$h_i = \mathrm{LSTM}_\phi(X_i) \ \in \ \mathbb{R}^h.$$

If we have an exogenous control signal C_i, we concatenate it to the LSTM's output:

$$v_i = \begin{cases} \left[h_i;\ C_i\right] \ \in \ \mathbb{R}^{h+1}, & \text{(conditional)} \\ h_i \ \in \ \mathbb{R}^h, & \text{(non-conditional)} \end{cases}$$

Finally, this vector is projected into the k-dimensional latent space by a linear layer (the encoder's final affine mapping):

$$z_i = W_\phi\, v_i \ + \ b_\phi \ \in \ \mathbb{R}^k.$$

We enforce the empirical distribution $\{z_i\}$ to align with the prior $p(z) = \mathcal{N}(0, I)$ in the sense of entropically-regularized Wasserstein-2. Given a batch of size B (Tolstikhin et al., 2019).

Record on Wasserstein-2 and Sinkhorn

For two distributions p, q on $\mathbb{R}^k$, the Wasserstein-2 distance is defined by the Kantorovich transport problem:

$$W_2^2(p,q) = \min_{\Gamma \in \Pi(p,q)} \int_{\mathbb{R}^k \times \mathbb{R}^k} \|x-y\|^2 \, \mathrm{d}\Gamma(x,y),$$

where Π is the :

$$\Pi(p,q) = \left\{ \Gamma \geq 0 \mid \forall A,\ \Gamma(A \times \mathbb{R}^k) = p(A),\ \Gamma(\mathbb{R}^k \times A) = q(A) \right\}$$

To make this problem computable and differentiable, we add an entropic regularization (Sinkhorn) term (Cuturi, 2013) :

$$W_{2,\varepsilon}^2(p,q) = \min_{\Gamma \in \Pi(p,q)} \sum_{i,j} \Gamma_{ij} \|x_i - y_j\|^2 \ - \ \varepsilon H(\Gamma)$$

where $\Gamma = [\gamma_{ij}]$, $\forall \gamma_{ij} \geq 0$ and $\sum_{i,j} \gamma_{ij} = 1$ demonstration :

$$H(\Gamma) = -\sum_{i,j} \Gamma_{ij} \log \Gamma_{ij} = -\sum_{i,j} \gamma_{ij} \log \gamma_{ij}, \quad \varepsilon > 0 \text{ (régularization).}$$

Discrete Sinkhorn-Wasserstein on a Batch

Given a batch of encoded latents $\{z_i\}_{i=1}^{B} \sim q_\phi(z)$ and B samples $\{z'_j\}_{j=1}^{B} \sim p(z) = \mathcal{N}(0, I)$. We can define the cost matrix :

$$C_{ij} = \| z_i - z'_j \|^2 \quad \text{for } i, j = 1, \dots, B$$

Enforce uniform marginals

$$\Gamma \mathbf{1}_B = \frac{1}{B} \mathbf{1}_B, \quad \Gamma^T \mathbf{1}_B = \frac{1}{B} \mathbf{1}_B$$

Entropically-regularized OT problem:

$$W_{\varepsilon,2}^2 = \min_{\Gamma \in \mathbb{R}_+^{B \times B}} \sum_{i,j} \Gamma_{ij} C_{ij} - \varepsilon \sum_{i,j} \Gamma_{ij} \log \Gamma_{ij}$$

Sinkhorn–Knopp Algorithm

Algorithm 1: Sinkhorn–Knopp Iterative Scaling Algorithm

Input: Cost matrix $C \in \mathbb{R}^{B \times B}$, regularization $\varepsilon > 0$, iterations T
Output: Regularized transport plan Γ and Sinkhorn cost $W^2_{\varepsilon,2}$

Compute the Gibbs kernel demonstration:

$$K_{ij} = \exp(-C_{ij}/\varepsilon)$$

Initialize scaling vectors:

$$u^{(0)} = v^{(0)} = \mathbf{1}_B$$

for $t = 0$ **to** $T - 1$ **do**

$$u^{(t+1)} = a \oslash (K\, v^{(t)}), \qquad v^{(t+1)} = b \oslash (K^T\, u^{(t+1)})$$

Form transport plan:

$$\Gamma = \operatorname{diag}(u^{(T)})\, K\, \operatorname{diag}(v^{(T)})$$

Compute Sinkhorn cost:

$$W^2_{\varepsilon,2} = \sum_{i,j} \Gamma_{ij}\, C_{ij}$$

Regularization Term in WAE–Sinkhorn Loss

The latent vector z_i (and C_i if conditional) is then repeated L times and fed into the decoder:

$$\tilde{v}_i = \begin{cases} \left[z_i;\ C_i\right] \in \mathbb{R}^{k+1}, & \text{(conditional)} \\ z_i \in \mathbb{R}^{k}, & \text{(non-conditional)} \end{cases} \qquad U_i = \mathrm{Repeat}\Big(\tilde{v}_i,\ L\Big) \in \mathbb{R}^{L \times \text{dim}},$$

$$\widehat{X}_i = \mathrm{LSTM}_\theta\Big(U_i\Big), \qquad \widehat{X}_i(t) = W'_\theta \left[\mathrm{LSTM\_hidden}_t\right] + b'_\theta.$$

The total function loss is :

$$\mathcal{L} = \underbrace{\frac{1}{B}\sum_{i=1}^{B}\left\|X_i - \widehat{X}_i\right\|^2}_{\text{Reconstruction MSE}} + \underbrace{\lambda\, W^2_{\varepsilon,2}\Big(q_\phi(z),\ p(z)\Big)}_{\text{Sinkhorn OT penalty}}$$

- $W^2_{\varepsilon,2}(q,p) = \min\limits_{\Gamma\in\Pi_B} \sum\limits_{i,j} \Gamma_{ij}\,\|z_i - z'_j\|^2 \; - \; \varepsilon\, H(\Gamma)$
- $\Pi_B = \{\Gamma \geq 0 : \Gamma\mathbf{1} = \frac{1}{B}\mathbf{1},\ \Gamma^T\mathbf{1} = \frac{1}{B}\mathbf{1}\}$, in practice: $\lambda = 1$.

Remarks:

- $\varepsilon \to 0 \Rightarrow$ approaches true W_2^2, but may cause numerical instability.
- ε large $\Rightarrow$ smoother transport plan, faster convergence, but less geometric fidelity.

New vision for errors in econometrics

Stress as a latent dimension.

If one adopts the reconstruction error e_i as a proxy for the level of financial stress at time i, then:

$$\mathrm{Stress}_i \;\equiv\; e_i \quad \left(\text{où } e_i = \frac{1}{Ld} \sum_{t=1}^{L} \sum_{j=1}^{d} |X_i(t,j) - \widehat{X}_i(t,j)| \right)$$

Two zones can be defined: calm zone $\mathrm{Stress}_i \leq T_\alpha$ and High stress zone: $\mathrm{Stress}_i > T_\alpha$.

- $\mathrm{Stress}_i^{\mathrm{nc}}$ (blue): "idiosyncratic" stress without Forward Guidance (FG), the hypothetical stress level had no FG signal been issued.
- $\mathrm{Stress}_i^{\mathrm{c}}$ (orange): "residual" stress after accounting for FG
- $\Delta_i = \mathrm{Stress}_i^{\mathrm{nc}} - \mathrm{Stress}_i^{\mathrm{c}}$, the quantity Δ_i can be interpreted as the marginal contribution of Forward Guidance (FG) to systemic stress reduction at time i. It captures the portion of financial stress that can be attributed to the absence of FG, by comparing the latent stress level without FG to the residual stress after FG has been accounted for.

MRE
MONTPELLIER
RECHERCHE
ECONOMIE

Model Comparison: FG Scores via Ridge Regression

Embedding Model	Input Type	FG Score Dim.	R^2 (Ridge)
Mamba (130M)	Sentence mean pooling	$\mathbb{R}^{768}$	**0.77**
Llamba (7B)	Sentence mean pooling	$\mathbb{R}^{4096}$	0.68
ModernBERT	Sentence mean pooling	$\mathbb{R}^{768}$	0.58

Best performance with Mamba embeddings for predicting systemic stress.

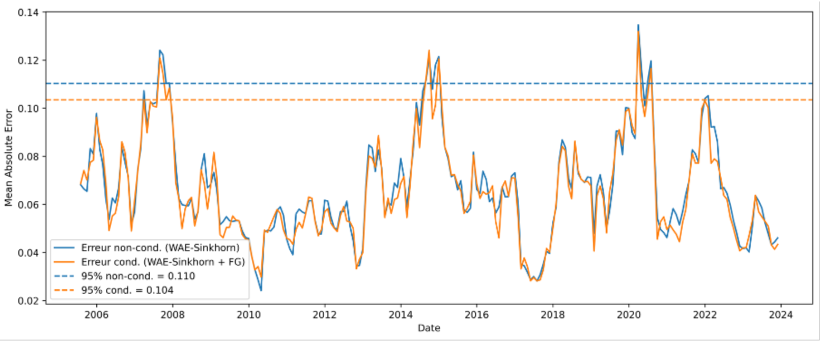

Figure: Superimposed reconstruction errors of the conditional and non-conditional models (short term).

In the short term (7-day horizon), latent stress is generally well anticipated, and Forward Guidance replicates systemic stress without notable attenuation.

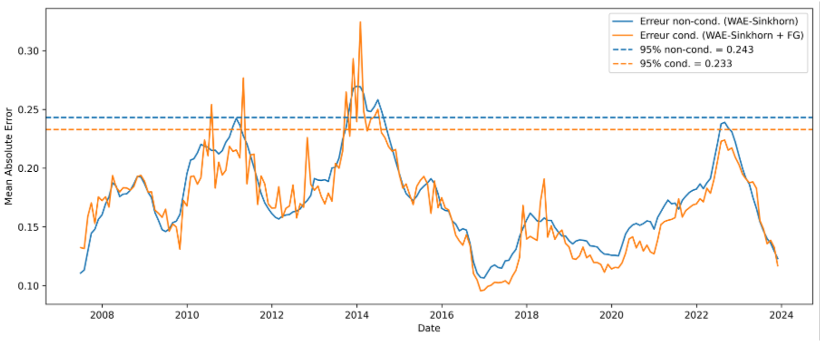

Figure: Superimposed reconstruction errors of the conditional and non-conditional models (mid term).

At the medium-term horizon (30 days), the conditional model with Forward Guidance shows that during crisis periods, communication becomes erratic, with higher latent stress.

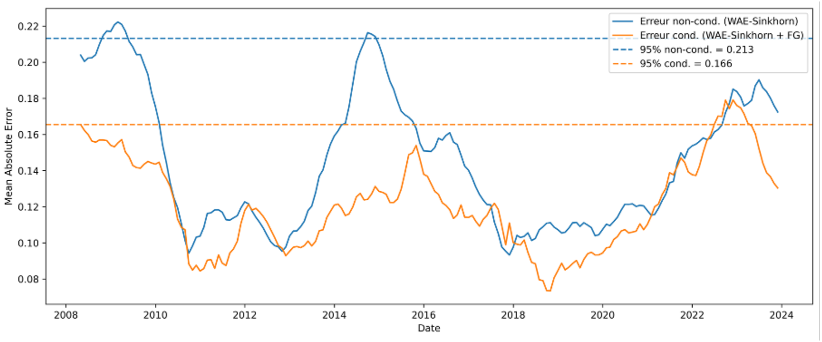

Figure: Superimposed reconstruction errors of the conditional and non-conditional models (long term).

In the long term (6 months), a communication lag of several months can be observed between the ECB and systemic stress.

Perspectives & Discussion

- **Alternative OT divergences:**
  - Sinkhorn divergences (subtract self-transport terms) for unbiased estimates.
  - Unbalanced OT to handle batch size mismatches or missing data.
- **Richer decoders / temporal models:**
  - Replace LSTM with Transformer or Temporal Convolution for faster parallel training.
  - Add attention over latent+control for improved sequence reconstruction.
- **Exogenous signals & multimodality:**
  - Incorporate additional macro indicators (CPI, unemployment, market liquidity).
  - Multimodal extension: combine text-based FG embeddings with numeric data.

THANK YOU FOR YOUR ATTENTION !

References II

Dao, T. and Gu, A. (2024). Transformers are ssms: Generalized models and efficient algorithms through structured state space duality.

Gu, A. and Dao, T. (2024). Mamba: Linear-time sequence modeling with selective state spaces.

Tolstikhin, I., Bousquet, O., Gelly, S., and Schoelkopf, B. (2019). Wasserstein auto-encoders.

Wang, J., Paliotta, D., May, A., Rush, A. M., and Dao, T. (2025). The mamba in the llama: Distilling and accelerating hybrid models.

Warner, B., Chaffin, A., Clavié, B., Weller, O., Hallström, O., Taghadouini, S., Gallagher, A., Biswas, R., Ladhak, F., Aarsen, T., Cooper, N., Adams, G., Howard, J., and Poli, I. (2024). Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference.

MONTPELLIER
RECHERCHE
ECONOMIE

Selective State Space Model Schema

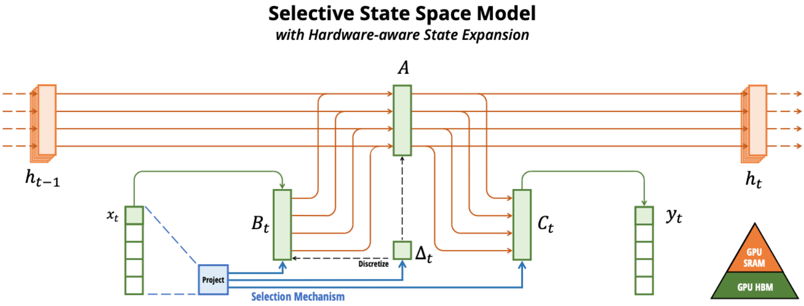

Figure: Selective State Space Model Mechanism.

Proof : Relative Entropy back

Let $\gamma \in \Pi(p, q)$ and $\mu = p \otimes q$, with $\gamma \ll \mu$. By Radon–Nikodym there exists

$$f(x, y) \; = \; \frac{d\gamma}{d\mu}(x, y), \quad \int h \, d\gamma \; = \; \int h \, f \, d\mu.$$

The Kullback–Leibler divergence is

$$\mathrm{KL}(\gamma \| \mu) = \int \log\big(f(x, y)\big) \, d\gamma(x, y),$$

and the *relative entropy* (negative KL) becomes

$$H(\gamma \| \mu) \; := \; -\mathrm{KL}(\gamma \| \mu) \; = \; - \int \log\big(f(x, y)\big) \, d\gamma(x, y)$$

In particular, with $\mu = p \otimes q$ on $\mathbb{R}^d \times \mathbb{R}^d$, $H(\gamma \| \mu)$ measures how far γ deviates from independent marginals.

$$H(\gamma) = - \int \log \left(\frac{d\gamma}{d(p \otimes q)}(x, y) \right) \, d\gamma(x, y)$$

Proof : Discretisation of Relative Entropy

Let:

- $p = (a_i)_{i=1}^n$ be a discrete probability distribution over the points $\{x_i\} \subset \mathbb{R}^d$,
- $q = (b_j)_{j=1}^m$ be a discrete probability distribution over the points $\{y_j\} \subset \mathbb{R}^d$,
- and $\gamma = (\gamma_{ij}) \in \mathbb{R}^{n \times m}$, a discrete transport plan between p and q.

Then:

$$p \otimes q = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i b_j \, \delta_{(x_i, y_j)} \quad \text{and} \quad \gamma = \sum_{i=1}^{n} \sum_{j=1}^{m} \gamma_{ij} \, \delta_{(x_i, y_j)}$$

At each atomic point (x_i, y_j), the Radon–Nikodym derivative (i.e., the density) of γ with respect to $p \otimes q$ is given by:

$$\frac{d\gamma}{d(p \otimes q)}(x_i, y_j) = \frac{\gamma_{ij}}{a_i b_j} \quad (\text{since } (p \otimes q)(x_i, y_j) = a_i b_j)$$

The continuous formula thus becomes:

$$H(\gamma) = -\sum_{i,j} \gamma_{ij} \log \left(\frac{\gamma_{ij}}{a_i b_j} \right)$$

Proof : Discretisation of Relative Entropy

We expand:

$$H(\gamma) = -\sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_{i,j} \gamma_{ij} \log(a_i b_j) = -\sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_{i} \log a_i \sum_{j} \gamma_{ij} + \sum_{j} \log b_j \sum_{i} \gamma_{ij}$$

Using the marginal constraints: $\sum_{j} \gamma_{ij} = a_i, \quad \sum_{i} \gamma_{ij} = b_j$, we obtain:

$$\sum_{i} \log a_i \sum_{j} \gamma_{ij} = \sum_{i} \log a_i \cdot a_i = \sum_{i} a_i \log a_i, \quad \sum_{j} \log b_j \sum_{i} \gamma_{ij} = \sum_{j} \log b_j \cdot b_j = \sum_{j} b_j \log b_j$$

Therefore:

$$H(\gamma) = -\sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_{i} a_i \log a_i + \sum_{j} b_j \log b_j$$

Since the terms $\sum_i a_i \log a_i$ and $\sum_j b_j \log b_j$ are constant when optimizing over $\gamma \in \Pi(p, q)$, they can be ignored in the objective. We obtain the simplified discrete entropy form:

$$H(\Gamma) := -\sum_{i,j} \gamma_{ij} \log \gamma_{ij}$$

Proof : Wasserstein-Sinkhorn Algorithm back

We consider two discrete probability distributions $a = (a_i)_{i=1}^n$ and $b = (b_j)_{j=1}^m$ on $\mathbb{R}^d$, supported on points $\{x_i\}, \{y_j\}$ respectively. The entropically regularized Wasserstein-2 distance is defined as:

$$W_{2,\varepsilon}^2(a,b) = \min_{\Gamma \in \Pi(a,b)} \sum_{i,j} \gamma_{ij} \|x_i - y_j\|^2 - \varepsilon \sum_{i,j} \gamma_{ij} \log \gamma_{ij} \tag{1}$$

where $\Pi(a,b)$ is the set of couplings $\Gamma \in \mathbb{R}_+^{n \times m}$ satisfying: $\sum_j \gamma_{ij} = a_i, \quad \sum_i \gamma_{ij} = b_j$

We introduce Lagrange multipliers α_i and β_j to enforce the marginal constraints. The Lagrangian becomes:

$$\mathcal{L}(\Gamma, \alpha, \beta) = \sum_{i,j} \gamma_{ij} \|x_i - y_j\|^2 - \varepsilon \sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_i \alpha_i \left(a_i - \sum_j \gamma_{ij} \right) + \sum_j \beta_j \left(b_j - \sum_i \gamma_{ij} \right)$$

To find the optimal Γ^*, we compute the stationary point of $\mathcal{L}$ with respect to each γ_{ij}:

$$\frac{\partial \mathcal{L}}{\partial \gamma_{ij}} = \|x_i - y_j\|^2 - \varepsilon(1 + \log \gamma_{ij}) - \alpha_i - \beta_j = 0$$

Proof : Wasserstein-Sinkhorn Algorithm

Solving the equation:

$$\|x_i - y_j\|^2 - \varepsilon(1 + \log \gamma_{ij}) - \alpha_i - \beta_j = 0$$

leads to:

$$\log \gamma_{ij} = \frac{\alpha_i + \beta_j - \|x_i - y_j\|^2}{\varepsilon} - 1 \quad \Rightarrow \quad \gamma_{ij} = e^{-1} \cdot e^{\alpha_i/\varepsilon} \cdot e^{\beta_j/\varepsilon} \cdot e^{-\|x_i - y_j\|^2/\varepsilon}$$

Define:

$$u_i := e^{\alpha_i/\varepsilon}, \quad v_j := e^{\beta_j/\varepsilon}, \quad K_{ij} := \exp\left(-\frac{\|x_i - y_j\|^2}{\varepsilon}\right)$$

Then:

$$\boxed{\gamma_{ij} = u_i \cdot K_{ij} \cdot v_j} \quad \Rightarrow \quad \boxed{\Gamma = \mathrm{diag}(u)\, K\, \mathrm{diag}(v)}$$

Proof : Wasserstein-Sinkhorn Algorithm

We now impose the constraints $\Gamma\mathbf{1} = a$ and $\Gamma^\top\mathbf{1} = b$. Using the expression:

$$\Gamma = \mathrm{diag}(u)\, K\, \mathrm{diag}(v)$$

the constraints become:

$$\sum_j \gamma_{ij} = a_i \quad \Rightarrow \quad u_i(Kv)_i = a_i \quad \Rightarrow \quad u = \frac{a}{Kv}$$

$$\sum_i \gamma_{ij} = b_j \quad \Rightarrow \quad v_j(K^\top u)_j = b_j \quad \Rightarrow \quad v = \frac{b}{K^\top u}$$

This leads to the Sinkhorn-Knopp iterations :

$$\boxed{\begin{aligned} u^{(t+1)} &= \frac{a}{Kv^{(t)}} \\ v^{(t+1)} &= \frac{b}{K^\top u^{(t+1)}} \end{aligned}}$$

These iterations converge under mild conditions (e.g. $K_{ij} > 0$), providing the optimal Γ^*.